BARBERSHOP: INTEGRATED MONEY & BUSINESS MANAGEMENT SYSTEM

FYP4124 Final Year Project 2

Case study: KI'AD Barbershop, Ayer Hitam, Kedah | barbershophesmes.site.je

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PRESENTATION CONTENT

01 Introduction

02 Problem Statement & Objective

03 Methodology

04 System Design & Demo

05 Testing & Result

06 Future Works

07 Conclusion

01 - INTRODUCTION

Project Background

The Barbershop Management Gap

- Bookings taken by phone / WhatsApp, easy to double-book
- Cash and QR payments checked by hand, sales unclear
- Staff attendance, lateness and salary kept on paper
- Customer details scattered and not protected

The Solution: KI'AD Booking System

- One web platform: booking, payment, attendance and payroll
- Customer picks barber, date and time; clashes blocked
- Owner verifies QR receipts, tracks sales and staff pay
- Role-based login (customer, barber, owner), live on HTTPS

01 - INTRODUCTION

Target Users and Roles

Customer

- Register / log in with username or email
- Browse services and prices (RM8 - RM30)
- Book barber, date and time
- Pay cash or QR and upload receipt
- See My Bookings, receipts, rate the haircut

Barber (Staff)

- Log in to the STAFF panel
- See own bookings and schedule
- Mark payment cash received, mark Done
- Clock in / out by scanning the counter QR
- See own attendance and salary

Owner (Admin)

- Dashboard: sales, schedule, top services
- Verify or reject QR payments, reassign, cancel
- Attendance, performance ranking, payroll
- Manage services, staff, customers
- Shop settings: QR, hours, holidays, logo

02 - PROBLEM STATEMENT & OBJECTIVE

From Problems to Objectives

Problem Statement	Objectives
01 Bookings by phone / WhatsApp cause double bookings and missed slots	01 Develop online booking with barber choice and live slot availability that prevents double booking
02 Cash and QR payments are checked manually; sales and payments are unclear	02 Record cash / QR payments with owner verification and show sales on a live dashboard
03 Staff attendance, pay and customer data are scattered and unprotected	03 Centralise attendance (QR), performance and payroll behind secure role-based access

03 - METHODOLOGY

Agile Methodology

- Iterative stages: Plan → Design → Develop → Test → Deploy → Review
- Requirements refined through feedback from the owner, barbers and customers
- Flexible when the owner asks for changes (e.g. QR attendance, payroll, ratings were added in later iterations)
- Each iteration delivered a working feature on the live site

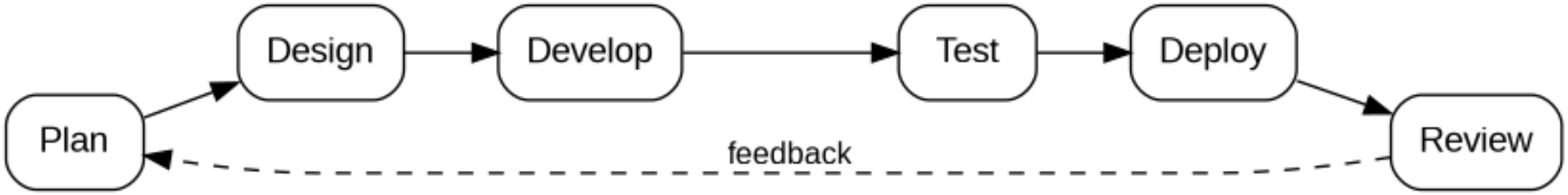


Figure 3.1: Agile Methodology Diagram

04 - SYSTEM DESIGN & DEMO

System Architecture



Front end: HTML, CSS, JavaScript, Bootstrap 5, Font Awesome | Back end: PHP 8, MySQLi | Hosting: shared hosting, HTTPS

04 - SYSTEM DESIGN & DEMO

Database Design (11 Tables)

Users

customer
username, email, password (hash)

staff
role: owner / staff, is_barber, active

Booking & Payment

receipt
date, time, barber, total, payment_method, payment_status, job_status

receipt_item
service name + price per booking

service
name, price, image

review
stars, comment, barber (1 per booking)

Staff Operations

attendance
clock_in / out, late_minutes

payroll
staff, month, amount, paid_at

Control

holiday
closed dates

setting
hours, slot length, QR, contact

login_attempt
wrong-password tracking

Relationships: receipt.cust_id → customer, receipt.barber_id → staff, receipt_item / review → receipt, attendance / payroll → staff

04 - SYSTEM DESIGN & DEMO

Booking to Payment Flow

1. **Choose services** — Search and add to cart
2. **Cart** — Review items and total
3. **Barber & time** — Any barber or a named one
4. **Payment** — Cash at shop or QR + receipt
5. **Owner checks** — Verify / reject QR receipt
6. **Service done** — Barber marks Done
7. **Review** — Customer rates 1-5 stars

Payment status

- cash: pay at the shop, owner/barber marks Cash received
- pending: QR receipt uploaded, waiting owner check
- verified: counted in 'Collected' sales
- rejected: chair released, customer must contact shop

Job status

- booked: appointment holds a chair
- done: barber finished, counted in barber sales
- cancelled: owner cancels, chair released
- Only 'done' bookings can be rated by the customer

04 - SYSTEM DESIGN & DEMO

Smart Booking Rules

1 Slot grid

30-minute slots from 10:00 to 22:00 (24 slots a day), set by the owner in Settings

2 No double booking

Same barber + same time is blocked when choosing the slot and re-checked at payment

3 Any barber

Auto-assigns the free barber with the fewest bookings that day

4 Closed days

Weekly closed days and holidays set by owner; past times are greyed out

5 Released chair

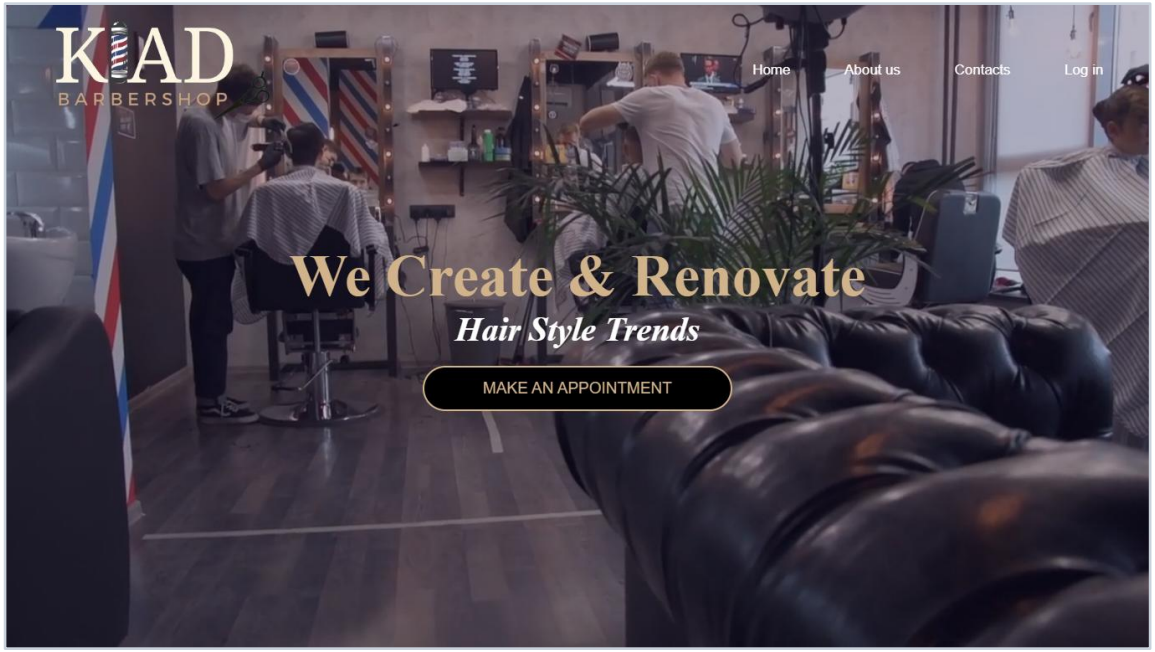
Rejected payments and cancelled bookings free the slot immediately

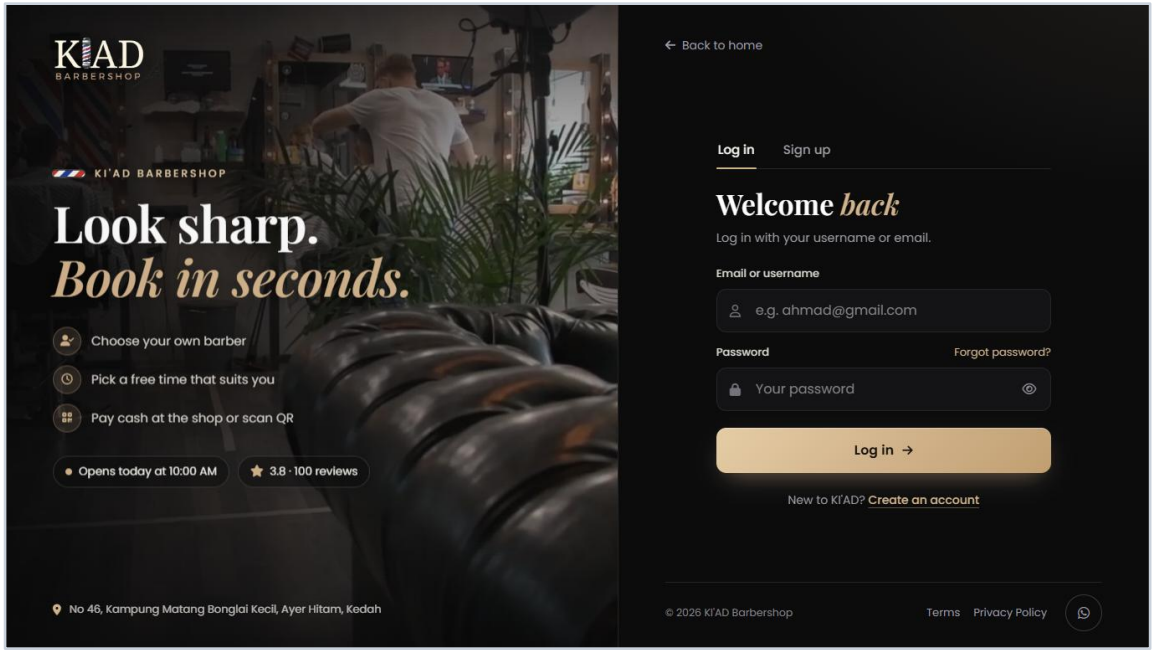
6 Named barber ranking

Bookings where the customer picks a barber by name are counted for performance

04 - SYSTEM DESIGN & DEMO

Demo: Public Website and Login





Home page

- Shop story, WhatsApp, map and contact
- Opening hours shown on the login page

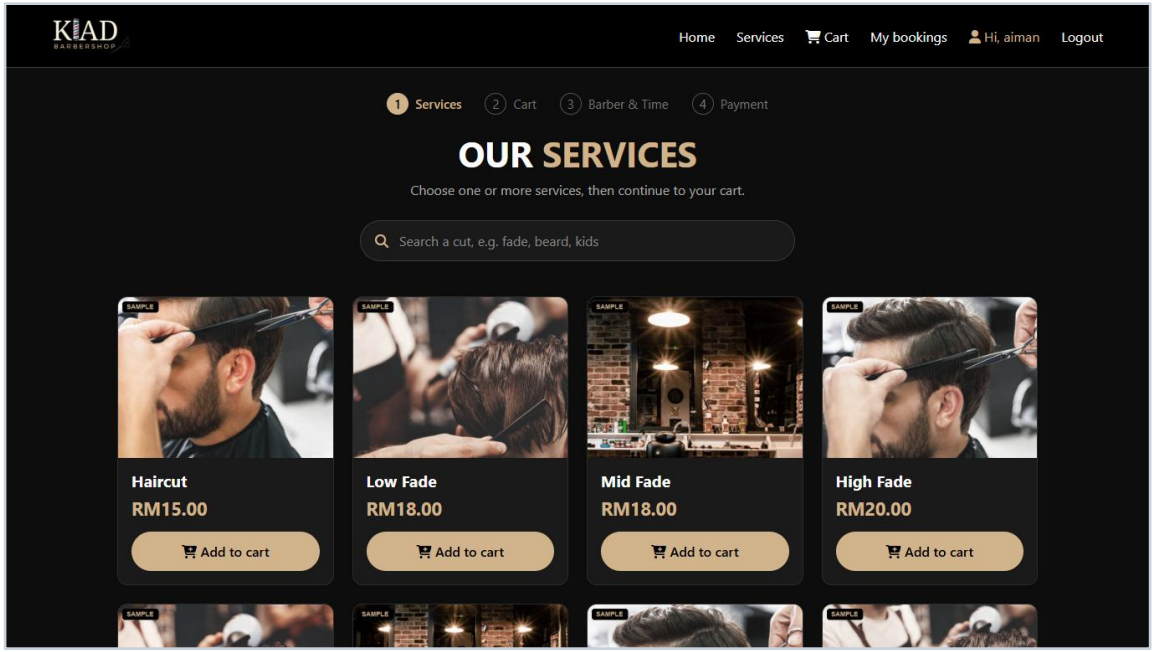
Log in / Sign up

- One login for customer, barber and owner
- Terms and Privacy page; password reset via WhatsApp to owner

Live site: https://barbershophesmes.site.je/home.php

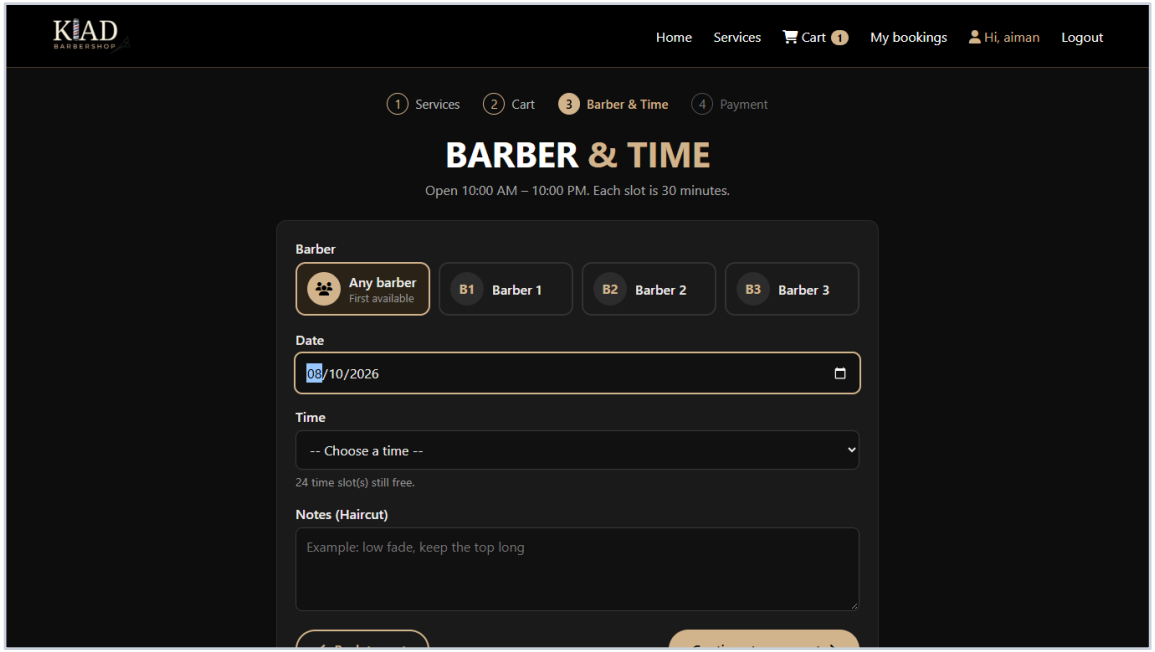
04 - SYSTEM DESIGN & DEMO

Demo: Customer Booking



Step 1: Services (search, add to cart)

- 9 services, RM8 - RM30, each with a photo
- Search by name or price, e.g. 'fade' or 'rm15'



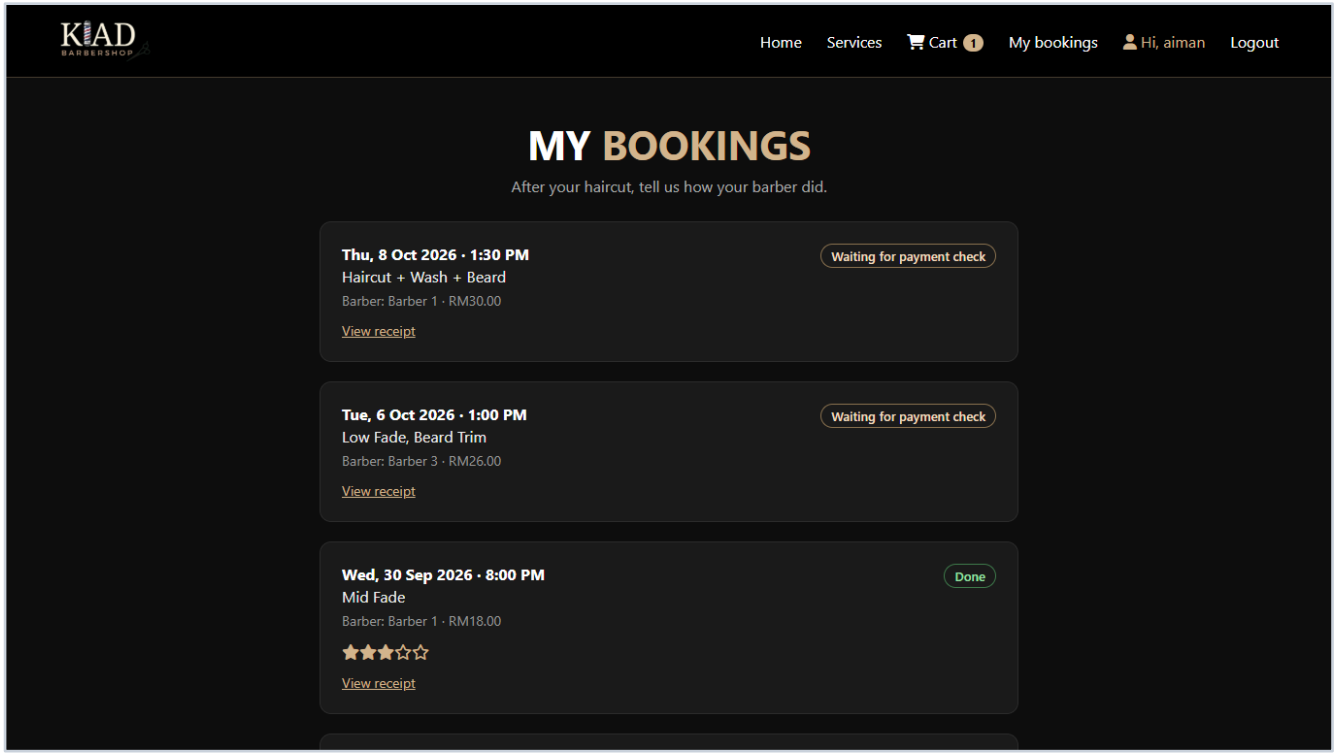
Step 3: Barber & time (live availability)

- Any barber (first available) or choose by name
- Notes for the haircut, then Continue to payment

Screenshots from a local copy with demo data

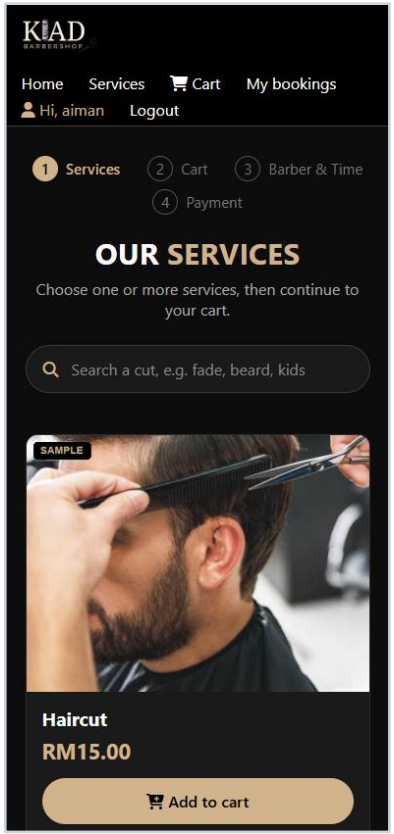
04 - SYSTEM DESIGN & DEMO

Demo: My Bookings and Mobile View



My Bookings: status, receipt, rating

- Status: Booked, Waiting for payment check, Payment rejected, Done
- Printable receipt; rate the barber after the haircut



Mobile view (390 px wide)

04 - SYSTEM DESIGN & DEMO

Demo: Owner Dashboard

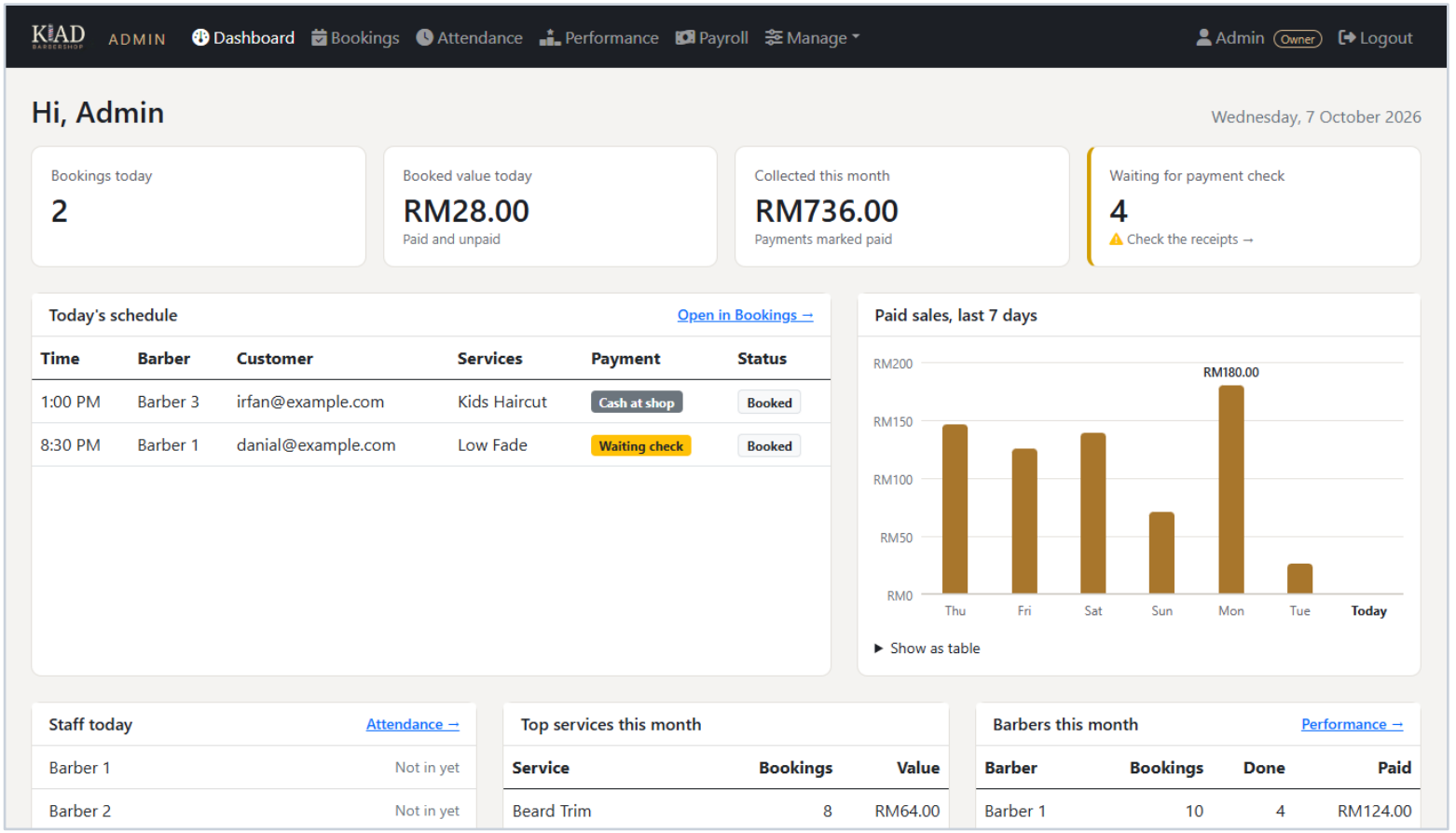


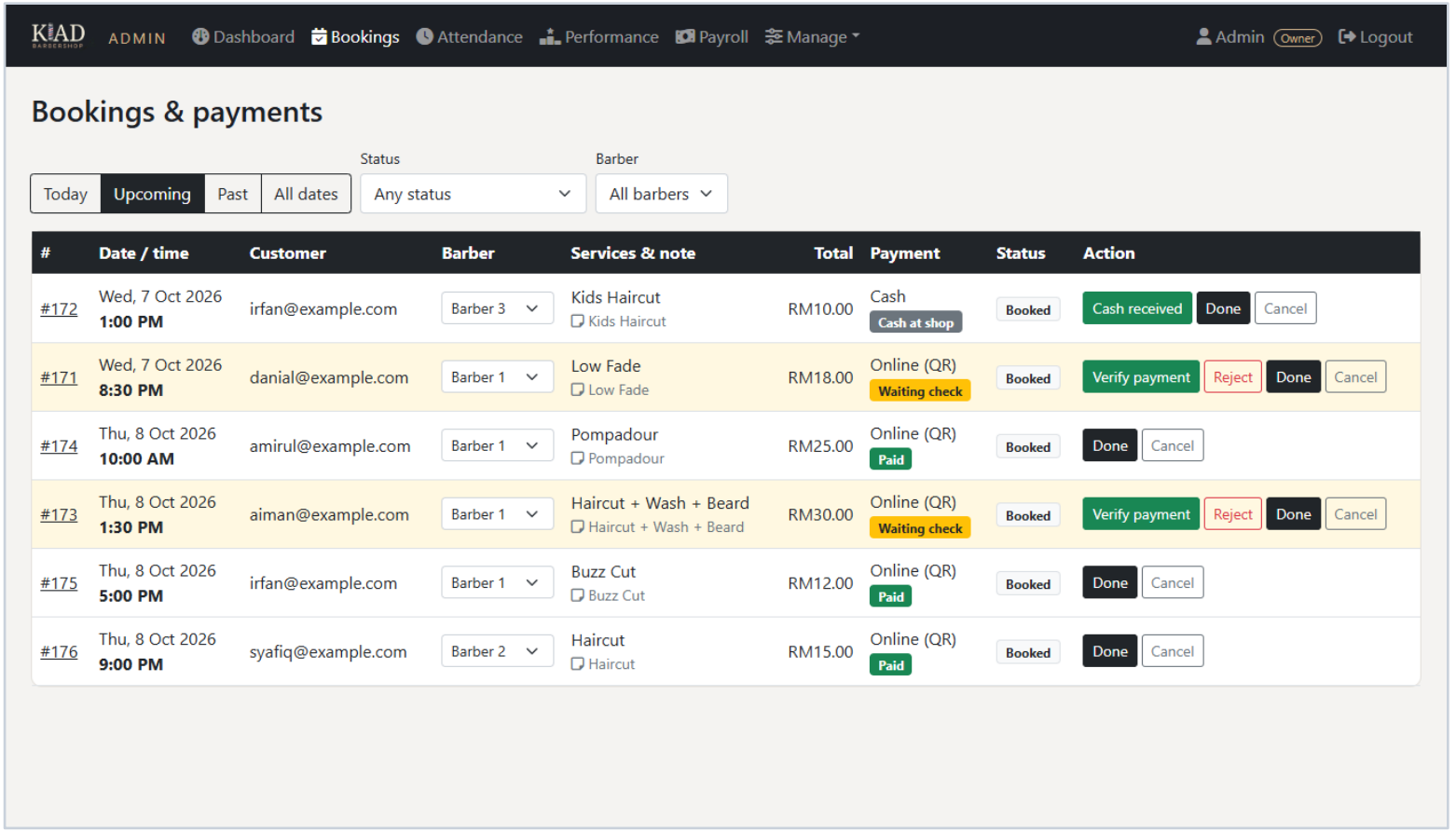
Figure 4.1: Owner dashboard (local demo data)

What the owner sees

- Bookings today and booked value
- Collected this month (verified payments only)
- Payments waiting for check
- Today's schedule per barber
- Paid sales, last 7 days (chart)
- Top services and barbers this month
- Who has clocked in today

04 - SYSTEM DESIGN & DEMO

Demo: Bookings & Payment Verification



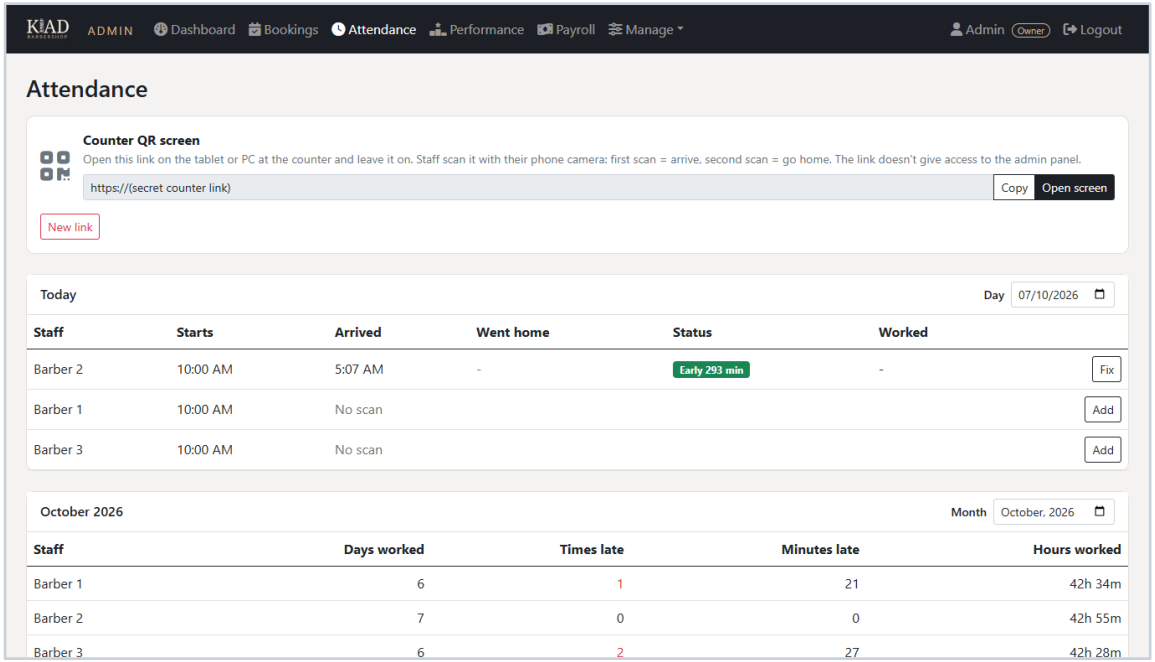
Actions

- **Verify / Reject** uploaded QR receipt (owner only)
- **Cash received** for pay-at-shop bookings
- **Done** when haircut finished
- **Reassign** barber from the row
- **Cancel** frees the slot
- Filters: Today, Upcoming, Past, status, barber

Figure 4.2: Bookings and payments page (local demo data)

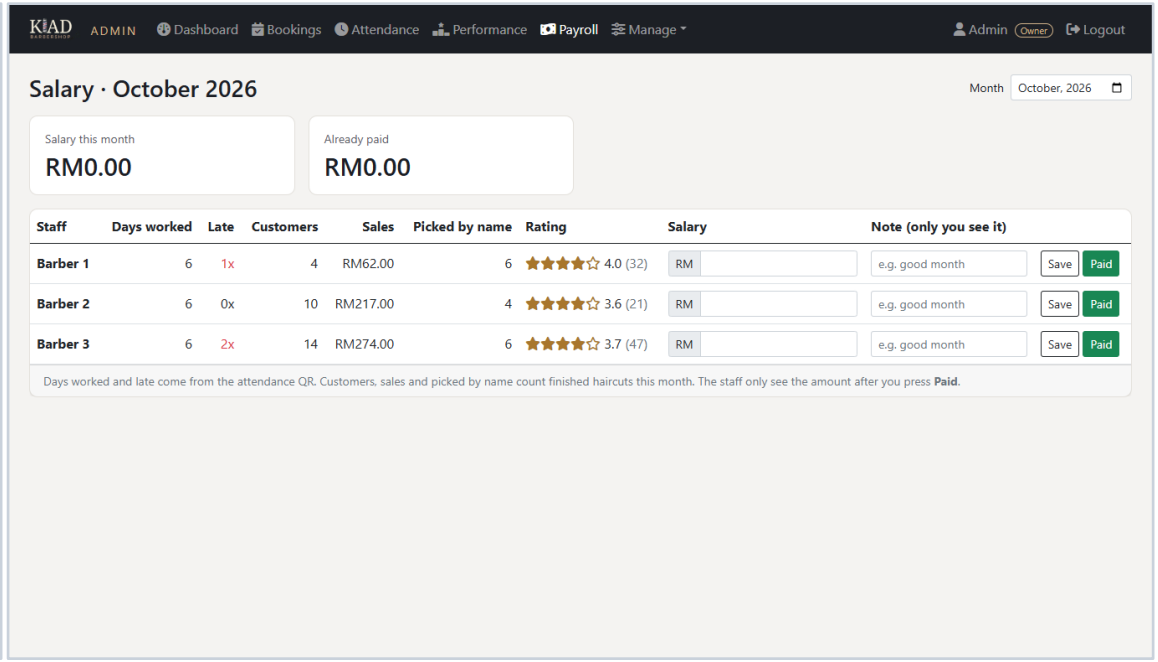
04 - SYSTEM DESIGN & DEMO

Demo: Attendance (QR) and Payroll



Attendance by counter QR

- First scan of the day = arrive, next scan = go home
- Late minutes counted after a 5-minute grace; owner can fix a forgotten scan

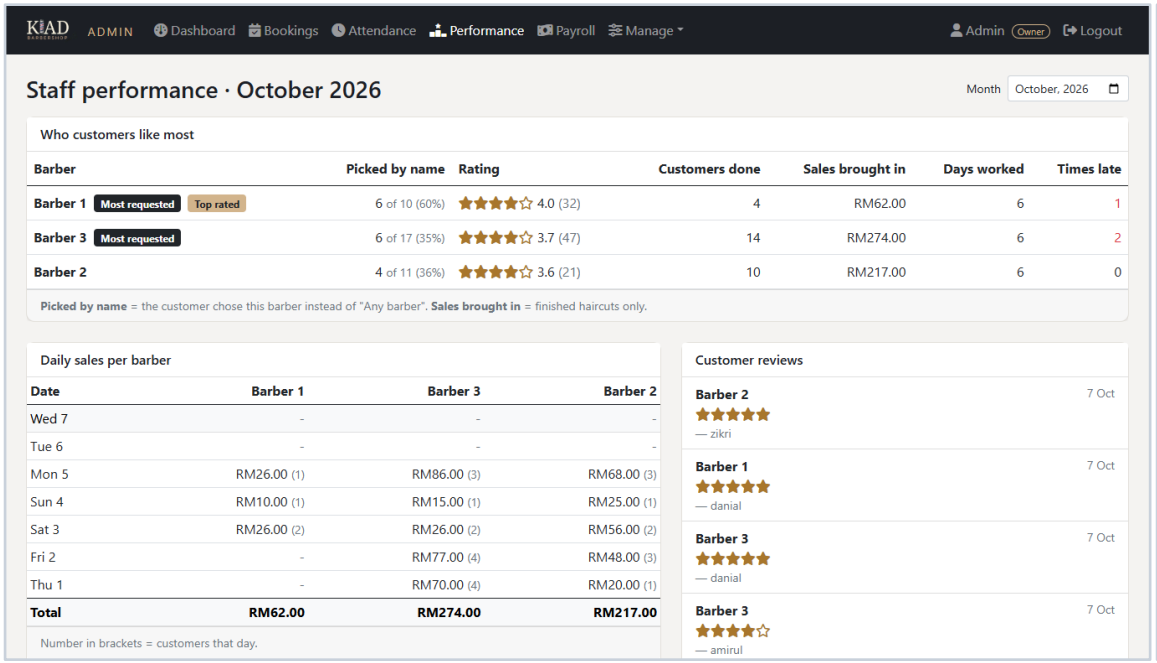


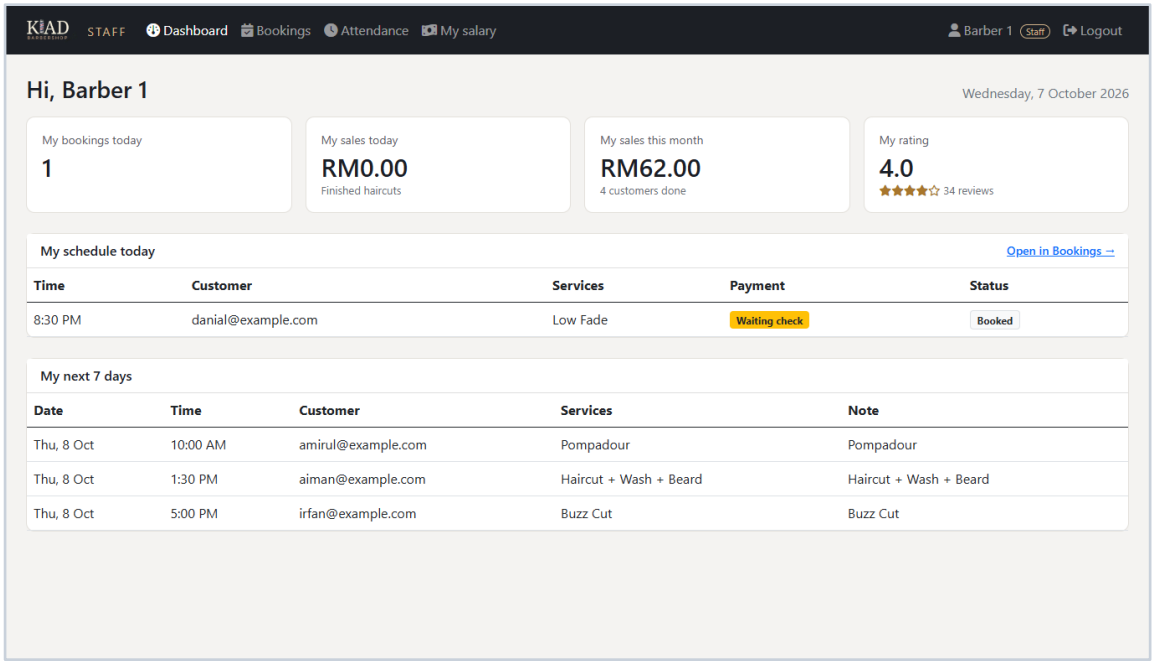
Monthly payroll

- Days worked, late, customers, sales, rating beside each salary
- A salary marked Paid is locked; staff see it only after Paid

04 - SYSTEM DESIGN & DEMO

Demo: Performance and Staff View





Performance (owner only)

- Customers served, sales, picked-by-name and star rating per barber
- Month picker for any past month

Barber (STAFF) dashboard

- Barber sees own bookings, sales and attendance only
- Owner-only pages are blocked for staff

04 - SYSTEM DESIGN & DEMO

Security Features Built In

1 Hashed passwords

bcrypt via password_hash; nobody at the shop can read a password

2 Login lockout

5 wrong tries per username (20 per IP) blocks login for 15 minutes

3 Role-based access

Customer, barber and owner pages are separated; owner-only actions re-checked on the server

4 CSRF + SQL injection

Token on every form; all queries use prepared statements

5 Safe sessions & HTTPS

HttpOnly, SameSite cookies, session ID regenerated at login, forced HTTPS in production, security headers

6 Protected files

config, SQL and logs blocked from the browser; receipts must be JPG / PNG / WEBP / PDF under 5 MB

05 - TESTING & RESULT

Non-Functional Testing: Performance

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-P01	Open the customer booking page (Services) and measure load time	Page loads within an acceptable time	Average 19 ms over 5 loads (max 30 ms)	**Pass**
NFT-P02	Open the owner dashboard with sales, schedule and chart data	Dashboard loads within an acceptable time	Average 104 ms over 5 loads (max 405 ms)	**Pass**
NFT-P03	Open the monthly payroll / performance page	Page is produced within an acceptable time	Average 22 ms over 5 loads (max 31 ms)	**Pass**

Table 4.1: Performance Testing Result Table (local server, 176+ bookings, 5 runs each)

05 - TESTING & RESULT

Non-Functional Testing: Security

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-S01	Log in with a wrong password	Login rejected with an error message	'Wrong username/email or password.' shown	**Pass**
NFT-S02	Customer opens an admin page by URL (payroll)	Access blocked; cannot view the page	HTTP 302 redirect to Login.php	**Pass**
NFT-S03	Staff opens owner-only Settings	Access denied for staff role	Redirected: 'Only the owner can open that page'	**Pass**
NFT-S04	Inspect stored passwords in the database	Passwords stored as hashes, not plain text	Staff and customer passwords start with $2y$ (bcrypt)	**Pass**
NFT-S05	Enter 6 wrong passwords for one account	Account login temporarily blocked	'Too many wrong passwords. Please wait 15 minutes'	**Pass**
NFT-S06	Submit the login form with a forged CSRF token	Request rejected	HTTP 400 returned	**Pass**
NFT-S07	Enter a SQL injection string as username	Login rejected, no data exposed	Rejected; prepared statements used	**Pass**
NFT-S08	Open config.php directly on the live website	Access forbidden	HTTP 403 Forbidden on barbershophesmes.site.je	**Pass**

Table 4.2: Security Testing Result Table

05 - TESTING & RESULT

Non-Functional Testing: Usability

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-U01	New customer completes a booking without help	Task completed without major confusion	*[Record result from user trial]*	
NFT-U02	Barber finds a booking and marks it Done	Menus and buttons easy to understand	*[Record result from user trial]*	
NFT-U03	Owner verifies a QR payment and checks sales	Form is clear and saves correctly	*[Record result from user trial]*	

Table 4.3: Usability Testing Result Table (to be completed with real users)

05 - TESTING & RESULT

Non-Functional Testing: Reliability

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-R01	Save a booking, log out, then log in again	Booking record is still stored	Booking still stored and listed in My Bookings	**Pass**
NFT-R02	Stop and restart the database and web server	All records remain accurate	176 seeded bookings and payment totals intact after restart	**Pass**
NFT-R03	Log in as customer, barber and owner one after another	Each role sees its own correct page	Home / STAFF panel / ADMIN panel respectively	**Pass**

Table 4.4: Reliability Testing Result Table

05 - TESTING & RESULT

Non-Functional Testing: Compatibility

Test ID	Test Description	Expected Result	Actual Result	Status
NFT-C01	Open the system on a desktop browser	Layout is readable and usable	Chrome, 1366 px: all pages render correctly	**Pass**
NFT-C02	Open the booking page on a mobile viewport	Responsive layout adapts to the screen	390 px viewport: single-column layout, no overflow	**Pass**

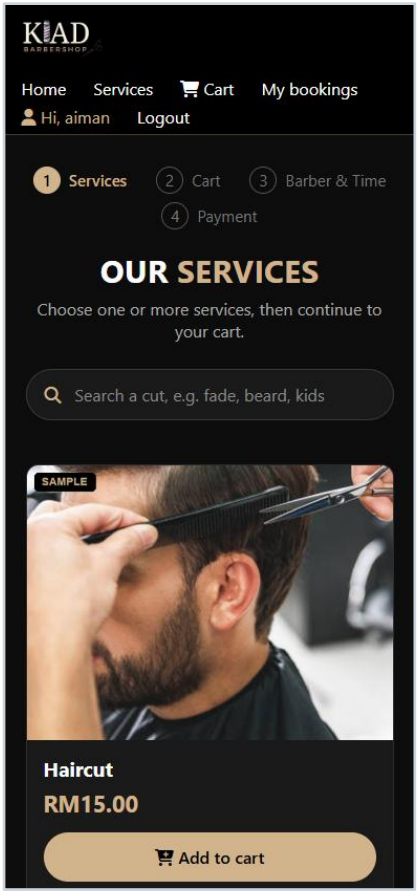


Table 4.5: Compatibility Testing Result Table (mobile run uses an emulated 390 x 844 viewport)

05 - TESTING & RESULT

Functional Testing: Unit Testing

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
UT01	Login	Enter valid username and password	Redirected to the correct page	Customer → home.php; owner → admin/index.php	**Pass**
UT02	Login	Enter wrong password	Login rejected with error message	'Wrong username/email or password.'	**Pass**
UT03	Booking	Select an available slot (cash)	Booking saved as booked, cash pending	Receipt created, payment_status = cash	**Pass**
UT04	Booking	Select an already booked slot	System prevents double booking	'Selected time slot is not available'	**Pass**
UT05	Payment	Mark a cash booking paid and done	Status saved in database	payment_status verified, job_status done	**Pass**
UT06	Sales	Compare dashboard sales with database	Total matches verified payments	RM736.00 on dashboard = SUM in DB	**Pass**
UT07	Access Control	Staff tries to verify an online payment	Action refused (owner only)	Payment status unchanged	**Pass**
UT08	Payment	Choose QR payment without a receipt	System asks for the receipt	'Please upload your payment receipt.'	**Pass**

Table 4.6: Unit Testing Result Table

05 - TESTING & RESULT

Functional Testing: Integration Testing

Test ID	Modules Involved	Test Scenario	Expected Result	Actual Result	Status
IT01	Booking + Barber assignment	Customer books 'any barber' at a busy time	A free barber is auto-assigned	Booked with free Barber 3; no clash	**Pass**
IT02	Booking + Payment	Barber marks cash received, then Done	Payment and job status update together	verified + done saved	**Pass**
IT03	Payment + Dashboard	Owner verifies a QR receipt	Counted in 'Collected' sales	payment_status = verified, in paid sales	**Pass**
IT04	Holiday + Booking	Owner marks a date closed; customer books it	Customer cannot book that date	'The shop is closed on this date'	**Pass**
IT05	User Role + Dashboard	Owner and barber log in	Dashboard shown by role	ADMIN menu for owner, STAFF for barber	**Pass**

Table 4.7: Integration Testing Result Table

05 - TESTING & RESULT

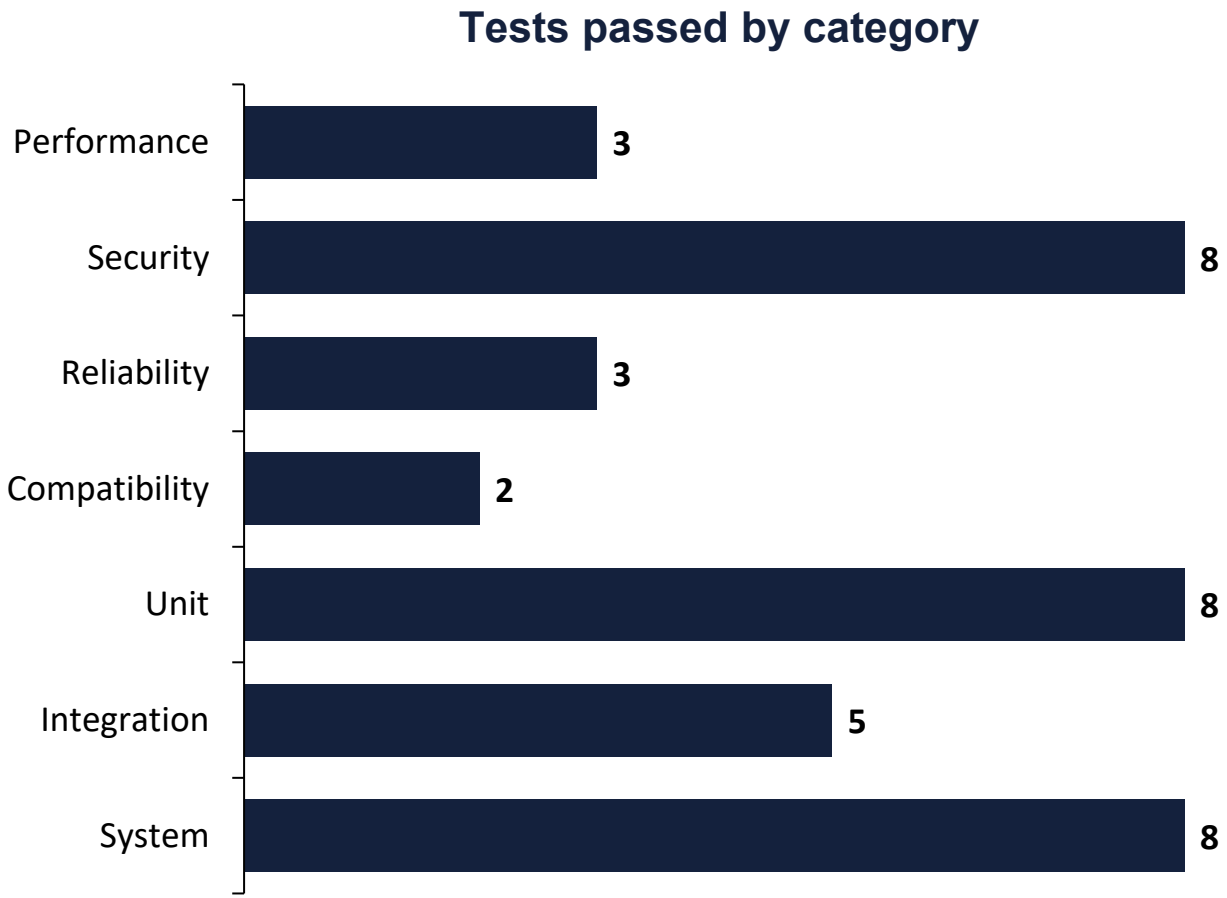
Functional Testing: System Testing

Test Scenario	Test Steps	Expected Result	Actual Result	Status
Customer registration and login	Sign up, then log in	Account created and login works	Registered; password hashed; login to home	Pass
Service and price viewing	Open the Services page	Name, photo and price shown	9 services, RM8 - RM30 shown	Pass
Appointment booking	Pick barber, date, slot, pay	Saved; double booking prevented	Saved; second booking on same slot refused	Pass
QR payment check	Upload receipt; owner verifies	Status moves pending → verified	verified and counted in sales	Pass
Attendance by QR	Barber scans the counter QR	Clock-in saved with late minutes	Attendance row created for the day	Pass
Counter QR protection	Open counter page with wrong key	Access refused	HTTP 403, no QR shown	Pass
Payroll	Owner saves salary and presses Paid	Saved, then locked	RM1500.00 saved; later edit ignored	Pass
Customer rating	Rate a finished haircut	Rating saved once per booking	5 stars saved; shown in Performance	Pass

Table 4.8: System Testing Result Table

05 - TESTING & RESULT

Test Summary



37

tests executed on the running system

37/37

passed (100%)

0

failed

Figure 4.3: Test results by category (usability trial pending)

05 - TESTING & RESULT

Acceptance Testing: User Feedback Questionnaire

Google Form: 15 questions | Likert scale 1 (strongly disagree) to 5 (strongly agree) | n = 27 responses

- 1. Role when trying the system
- 2. Register and log in is easy
- 3. Services, photos and prices are clear
- 4. Choosing barber, date and time is easy
- 5. Full slots are shown clearly
- 6. Payment process (cash / QR + receipt) is clear
- 7. Booking status and receipt are easy to understand
- 8. Pages load quickly
- 9. Easy to use on a mobile phone
- 10. Data and account feel safe
- 11. Dashboard, QR attendance, payroll help daily work (staff / owner)
- 12. Better than the old way at avoiding double bookings
- 13. Overall satisfaction
- 14. Would recommend the system (Yes / Maybe / No)
- 15. One thing to improve or add (open answer)

Table 4.11: User Acceptance Testing Questionnaire Items

05 - TESTING & RESULT

Acceptance Testing: Google Form Responses

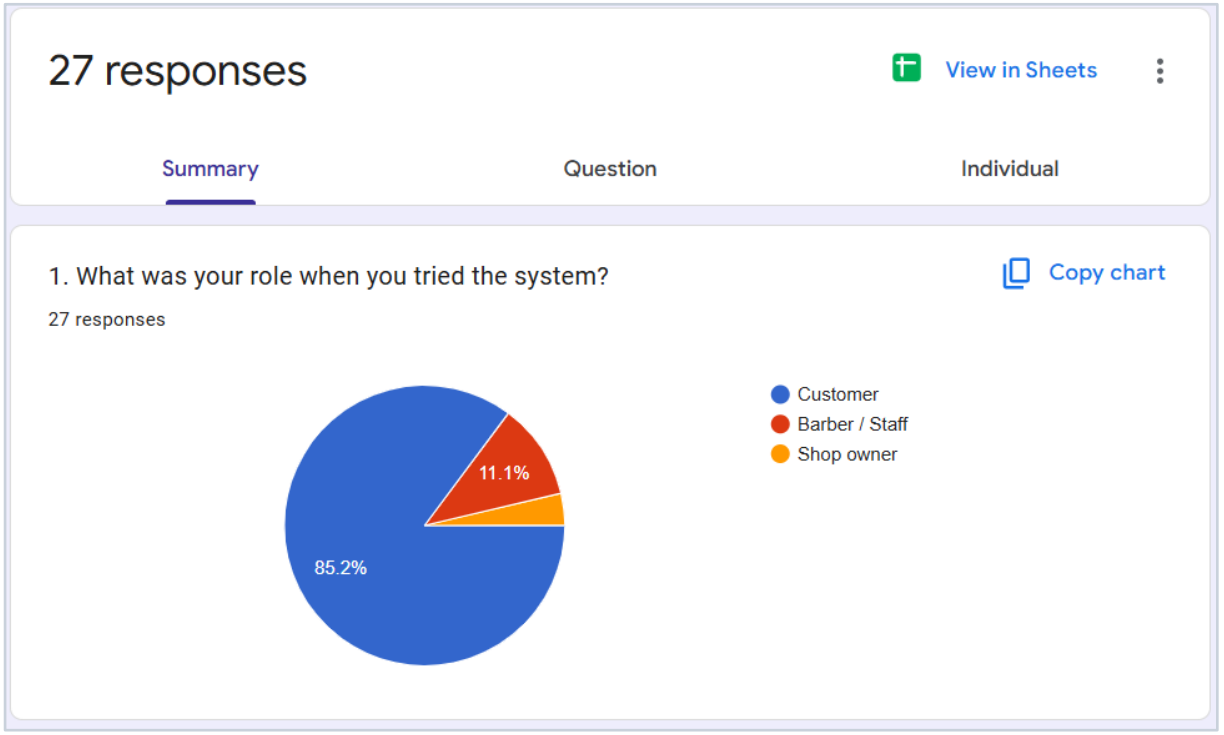


Figure 4.5: Google Forms summary

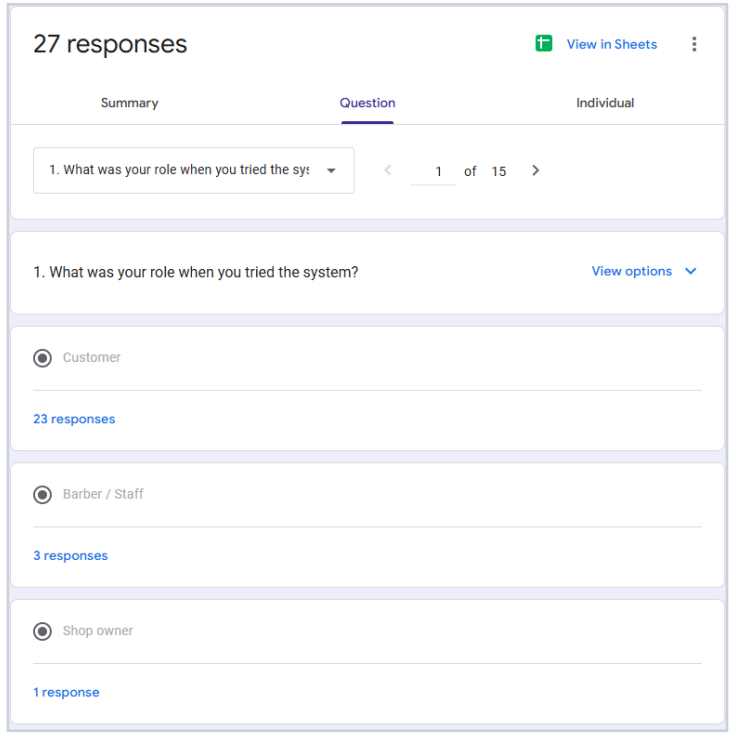


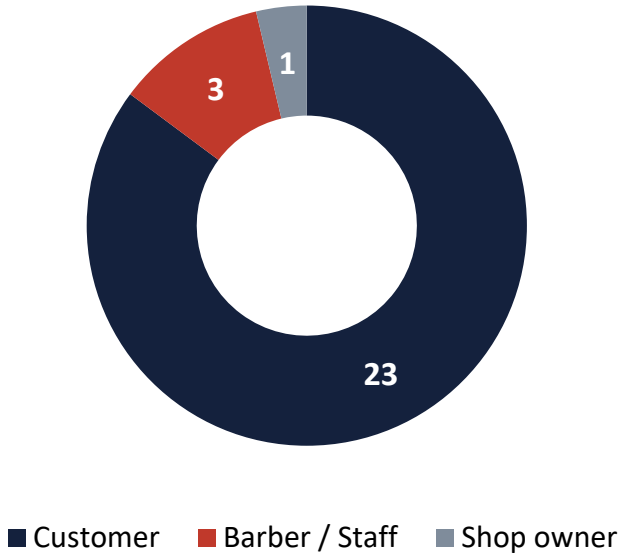
Figure 4.6: Responses per option (Question 1)

27 responses collected through Google Forms; the role split (23 / 3 / 1) matches the data analysed in the next slides.

05 - TESTING & RESULT

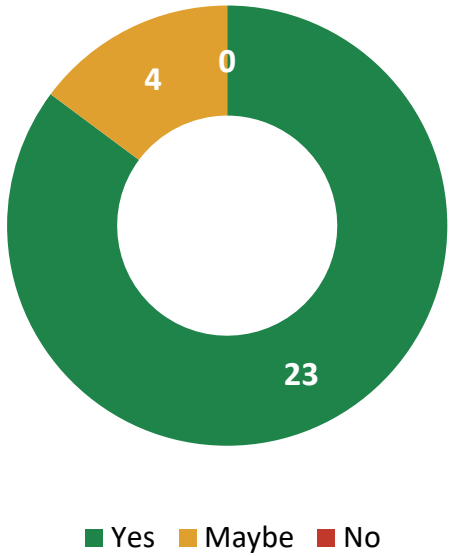
Acceptance Testing: Respondents and Recommendation

Figure 4.7: Respondent roles (n = 27)



23 customers, 3 barbers and 1 shop owner took part, so the results mainly reflect the customer experience.

Figure 4.8: Would recommend the system?



85% would recommend the system (23 of 27), 15% said maybe, and none said no.

05 - TESTING & RESULT

Acceptance Testing: Booking Experience

Figure 4.9: Mean score, booking steps (1-5)



Insight

- Highest: Register / log in (4.85), 100% agree
- Lowest: Services & prices (4.44), still above 4.4
- Average of the six items: 4.53 out of 5
- 15 of 162 answers were neutral or lower

Customers found booking easy overall. The lowest-rated item (Services & prices) is the best candidate for a small improvement.

05 - TESTING & RESULT

Acceptance Testing: Performance, Security and Overall

Figure 4.10: Mean score, quality and overall (1-5)



Insight

- 89% agree they are satisfied overall (mean 4.44)
- 96% say it beats the old phone / paper way (mean 4.7)
- 89% feel their data is safe
- Staff / owner (n = 4) rated the management pages 4.5 out of 5
- All 297 Likert answers: mean 4.54, 91% agree

Users rated the system 4.54 out of 5 on average. The strongest result is that it avoids double bookings better than the old way.

Note: question 11 counts barbers and the owner only; the sample is small (n = 27)

05 - TESTING & RESULT

Acceptance Testing: Client Testing & Result



Figure 4.11: Client Acceptance Testing Interview

- **Interviewee:** Mr Waqiyuddin, barbershop owner
- **Location & date:** [add shop name and date]
- **Purpose:** collect in-depth feedback from the owner
- Validate that the system meets business and operational needs
- Assess how booking, payment, attendance and payroll support daily management

05 - TESTING & RESULT

Acceptance Testing: Client Testing & Result

Question	System response	Outcome
Can the owner log in with a valid owner account?	Admin dashboard displayed	Login and role routing work
Can the owner see today's bookings?	Schedule lists time, barber, customer, service, payment	Suitable for daily operation
Can the owner verify a QR payment?	Receipt checked; status moves to Paid	Payment control achieved
Can services and prices be managed?	Service list edited with name, price and photo	Price transparency achieved

Table 4.9: Client Testing Checklist and System Response (1/2)

05 - TESTING & RESULT

Acceptance Testing: Client Testing & Result

Question	System response	Outcome
Can staff attendance be tracked?	Counter QR scan records arrival, lateness, departure	Attendance without paper
Can the owner prepare payroll?	Days worked, late, sales and rating shown beside salary	Supports fair salary decisions
Can sales and barber performance be seen?	Dashboard and Performance show paid sales and ratings	Supports owner decisions
Are admin pages protected from other users?	Customer and staff blocked from owner-only pages	Security requirement met

Table 4.10: Client Testing Checklist and System Response (2/2)

06 - FUTURE WORKS

1 Online Payment Gateway

Today the customer pays by QR and the owner checks the receipt. Next: automatic confirmation (FPX / e-wallet).

2 Expense & Profit Module

Record expenses (stock, rent) so the owner sees net profit, not only sales and salaries.

3 WhatsApp / SMS Reminders

Automatic booking reminders to reduce no-shows. Today the shop contacts customers by WhatsApp link.

4 Audit Log & 2FA

Record who changed what (payments, salaries) and add two-factor login for the owner.

5 Live Queue Display

Show waiting customers on a counter screen for walk-ins.

6 Loyalty & Mobile App

Reward repeat customers and offer an installable app (PWA).

07 - CONCLUSION

Built

A working booking, payment, attendance and payroll system for KI'AD Barbershop

Live

Deployed on HTTPS at barbershophesmes.site.je

Replaces paper

Phone bookings, manual payment checks and attendance books

Tested

37 tests on the running system, all passed

Prevents clashes

Double booking blocked at slot choice and at payment

Secure

Hashed passwords, lockout, CSRF, role-based access

PROJECT DEMO

Scan to try the KI'AD system

barbershophesmes.site.je

- Register as a customer and make a booking
- Owner and barber demo accounts are given on request
- Home, Services and Booking pages are public



THANK YOU

Questions are welcome